

Ishika Agarwal

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agarwalishika.github.io

Education

PhD, University of Illinois, Urbana-Champaign	Aug 2024 - May 2028 (expected)
- Specialization: Data Curation for NLP / Potential thesis title: "Autonomous Data Curation" - Advisor: Dr. Dilek Hakkani-Tür	
MS, University of Illinois, Urbana-Champaign	Aug 2022 - May 2024
Major: Computer Science	GPA: 3.88/4.00
Relevant courses: Applied ML, Info Retrieval, Generative Models, Computer Vision, ML+Data Systems, Adv. NLP	
BS, Purdue University	July 2019 - May 2022
Major: Computer Science (concentration in ML)	GPA: 3.88/4.00
Relevant courses: Data Mining and ML (Python), Web Information Search and Management (Python)	

Selected Research Publications

- [Under Review] **I. Agarwal**, A. Chakraborty, T. Sorensen, N. Gupta, A. Stolcke: LLMs Get Smarter with Targeted Synthetic Multilingual Data.
- [Under Review] **I. Agarwal***, S. Stoica*, E. Ackigoz, P. Natarajan, M. Namazifar, J. Ma, D. Hakkani-Tür: AcquisitionSynthesis: Targeted Data Generation using Acquisition Functions.
- [EMNLP 2026] **I. Agarwal**, Z. He, D. Patil, D. Hakkani-Tür: MTQE Rewards Implicitly Improve Translation Quality.
- [Preprint] **I. Agarwal***, N. B. Bozdag*, D. Hakkani-Tür: Language Specific Knowledge: Do Models Know Better in X Than in English?
- [NeurIPS 2025] **I. Agarwal**, D. Hakkani-Tür: Data Valuation with Neural Network for Efficient Instruction Fine-Tuning
- [ICLR 2025] **I. Agarwal**, K. Killamsetty, L. Popa, M. Danilevsky: DELIFT: Data Efficiency in Language model Fine-Tuning
- [EMNLP 2024] P. Kargupta*, **I. Agarwal***, D. Hakkani-Tür, J. Han: Instruct, Not Assist: LLM-based Multi-Turn Planning and Hierarchical Questioning for Socratic Code Debugging
- [ICLR 2024] Y.Ban, **I. Agarwal**, Z.Wu, Y.Zhu, K.Weldemariam, H.Tong, J.He: Neural Active Learning Beyond Bandits

Selected Experiences

Research Intern - Uniphore	May 2026 - Aug 2026
Researched data synthesis methods to improve a model's multilingual capabilities without degrading overall performance.	
Applied Science Intern - Amazon	May 2025 - Aug 2025
Interned in the Content Localization team at Prime Video. Researched data-efficient, multilingual idiom resolution.	
Research Scientist Intern - IBM Research	May 2024 - Aug 2024
Interned in the Conversational AI Model Development team. Developed new techniques to measure informativeness of data samples to efficiently fine-tune LLMs on instruction tuning data. Published DELIFT at ICLR 2024.	
Research Assistant - UIUC	August 2022 - May 2024
RA in Professor Hanghang Tong's IDEA Lab@UIUC. Working on three projects: (1) active graph anomaly detection (GAD), (2) efficient multi-armed bandits (MABs) for classification, and (3) an intersection between active GAD and efficient MABs.	
ML Engineering Intern - Apple	May 2023 - Aug 2023
Implemented a distributed training process for TensorFlow Ranker model used in the Apple Maps Search platform. Improved data generation and processing efficiency by 64%. Built feature engineering jobs in Spark.	
Software Engineer - Cisco WebEx	Feb 2022 - Aug 2022
Handled customer cases by debugging meeting issues and deploying fixes. Improved meeting features and mentored incoming summer interns. Trained coworkers in different teams on how to develop an internal debugging tool.	
Research Assistant - Purdue University	May 2021 - Feb 2022
RA in Professor Suresh Jagannathan's Lab, researching safe RL agents. Submitted to SafeAI @ AAAI 2021.	
Software Engineer Intern - Cisco WebEx	June 2021 - Aug 2021
Improved the internal logging and debugging tool for WebEx meetings which displays meeting records.	

Service

- Selected as **Top Reviewer for NeurIPS 2025**.
- Reviewer for NeurIPS 2024/2025, ICLR 2024/2025, AISTATS 2025/2026, ACL 2025, EMNLP 2024
- Teaching Assistant for (1) Advanced Information Retrieval and (2) Advanced NLP